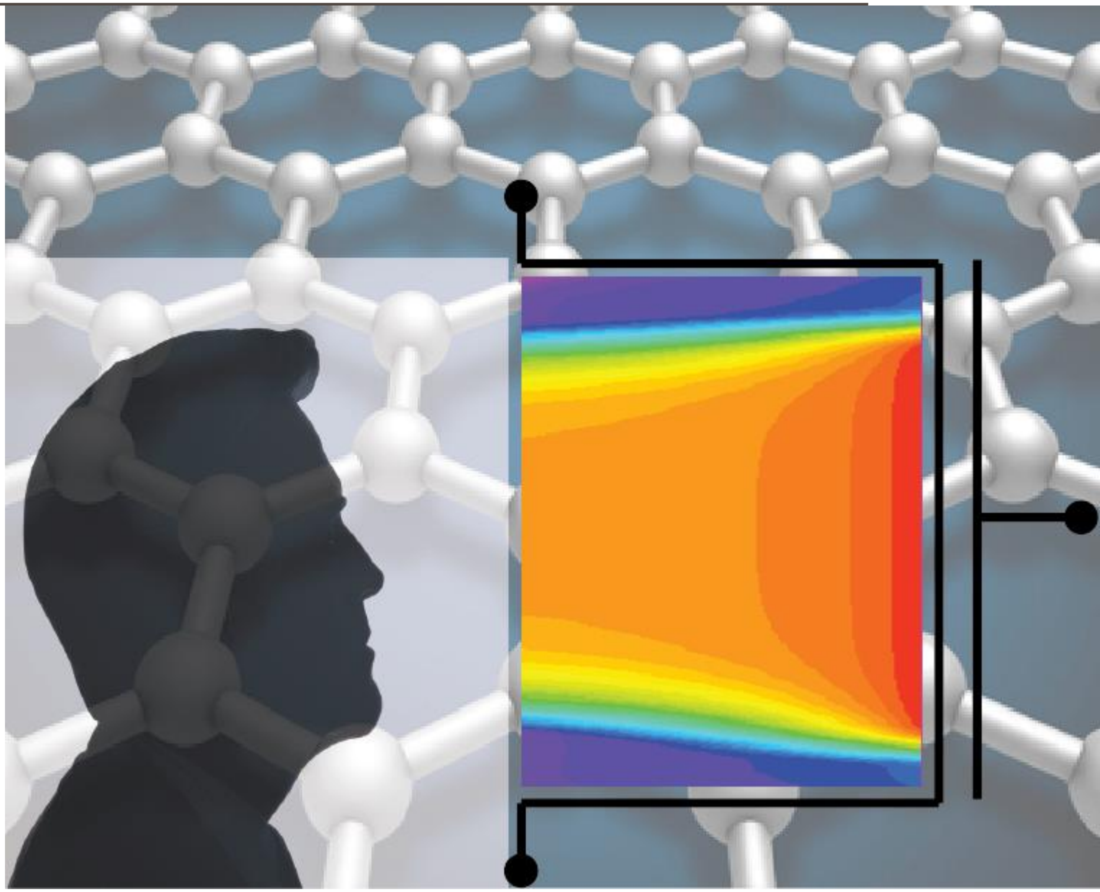
ADVANCED SOLID-STATE DEVICES

Week-3, Lecture-2



Sneh Saurabh
23rd January, 2025

Carrier Transport

Diffusion

Diffusion

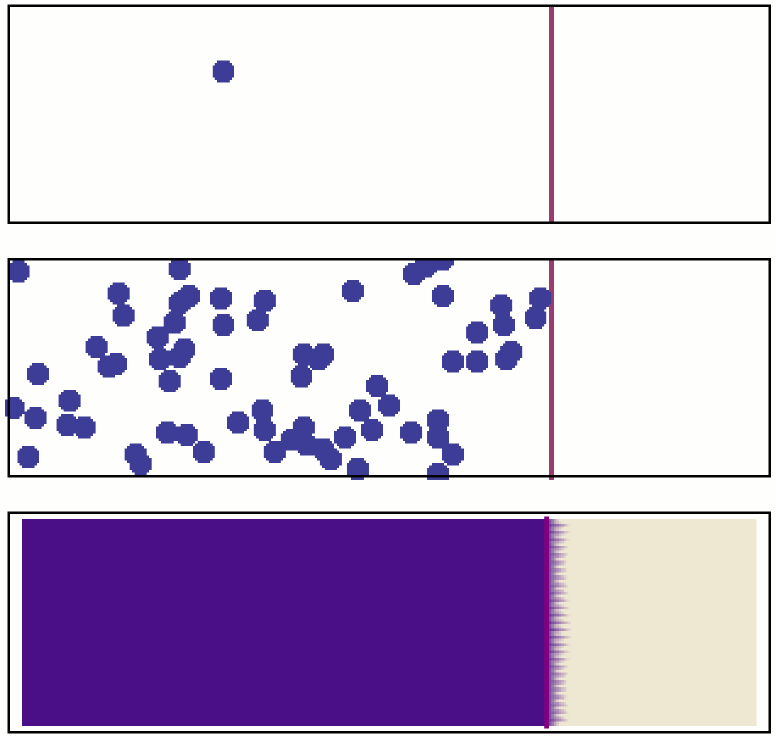
- Process by which particles/entities flow from a region of high concentration to a region of low concentration

- For example: diffusion of gas molecules

- Diffusion occurs due to random motion of particles/molecules

- Carrier transport by diffusion involves same process in which entities involved are electrons and holes

- It occurs due to **random thermal motion of carriers**



Diffusion Current Density (1)

Problem: Consider that the electron concentration varies as shown alongside:

$x = -l$, concentration = n_{-l}

$x = 0$, concentration = n_0

$x = +l$, concentration = n_{+l}

Assume that the thermal velocity of electrons v_{th} is same along X-axis (will be true if $2l$ is smaller than mean free path and no collision occurs).

Compute the net rate of electron flow (flux) F_n in +x direction at $x = 0$

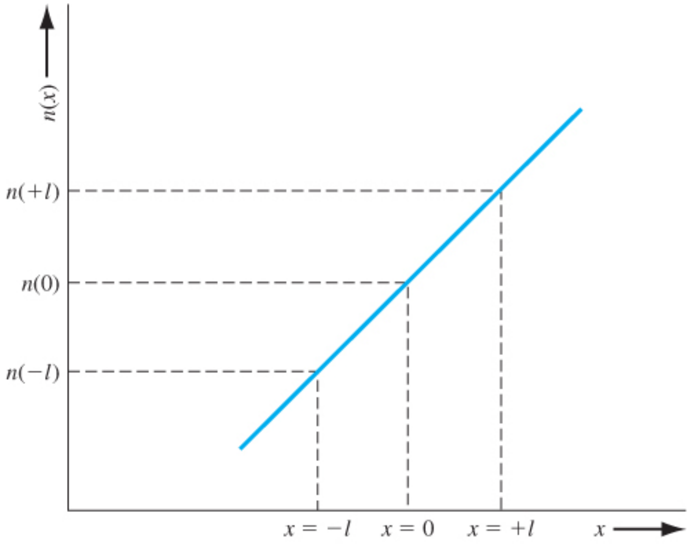


Figure 5.10 | Electron concentration versus distance.

Note: We consider the thermal velocity and not the drift velocity of carriers

Assume that: Due to random thermal motion, 50% of electrons will move along right and 50% along left at $x = -l$ and $x = +l$

$$Flux = \frac{Quantity}{Area \times Time}$$

Diffusion Current Density (2)

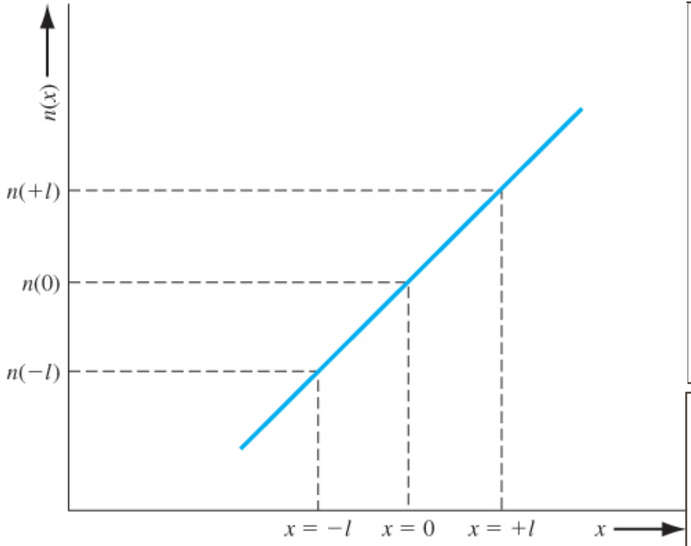


Figure 5.10 | Electron concentration versus distance.

- At $x = -l$, due to random thermal motion, 50% of electrons will move along right and 50% along left.

  => Flux towards right $= \frac{1}{2} n_{-l} v_{th}$

- These electrons will cross towards right at $x = 0$

- At $x = +l$, due to random thermal motion, 50% of electrons will move along right and 50% along left.

  => Flux towards left $= \frac{1}{2} n_{+l} v_{th}$

- These electrons will cross towards left at $x = 0$

Net flow of electrons towards right at $x = 0$,

$$F_n = \frac{1}{2} n_{-l} v_{th} - \frac{1}{2} n_{+l} v_{th} = \frac{1}{2} v_{th} (n_{-l} - n_{+l})$$

Diffusion Current Density (3)

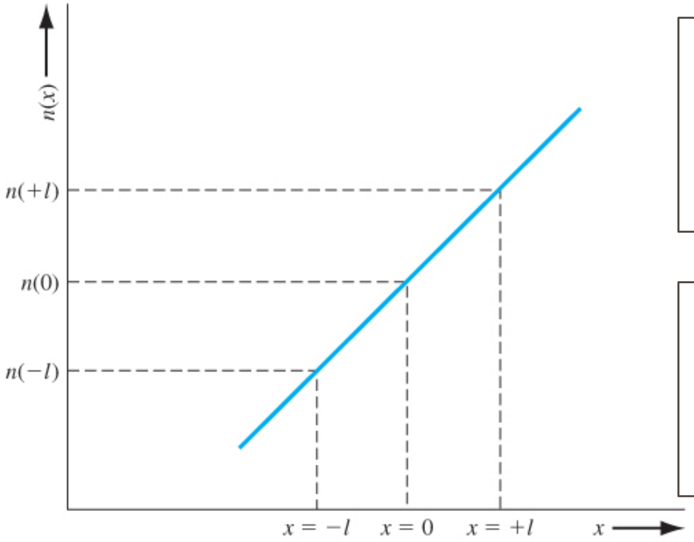


Figure 5.10 | Electron concentration versus distance.

Net flow of electrons towards right at $x = 0$,

$$F_n = \frac{1}{2} v_{th}(n_{-l} - n_{+l})$$

Taylor series expansion of n_x at $x = 0$:

$$F_n = \frac{1}{2} v_{th}\left(\left(n_0 - l\frac{dn}{dx}\right) - \left(n_0 + l\frac{dn}{dx}\right)\right) = -v_{th} l \frac{dn}{dx}$$

Each electron has charge: $-q$

Current density J is:

$$J = -qF_n = q v_{th} l \frac{dn}{dx}$$

Diffusion Coefficients

- Diffusion of carriers occur from high concentration region to low concentration region

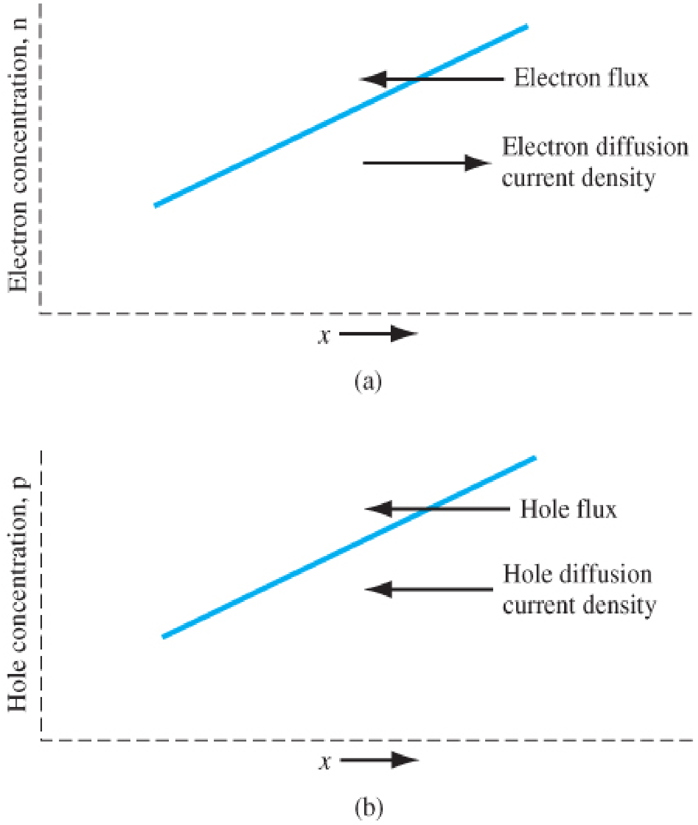


Figure 5.11 | (a) Diffusion of electrons due to a density gradient. (b) Diffusion of holes due to a density gradient.

Electron Diffusion Current: conventional current in the opposite direction (+x axis).

- We can also write the diffusion current as:

$$J_{nx,dif} = qD_n \frac{dn}{dx}$$

where D_n is the electron diffusion coefficient

Hole Diffusion Current: conventional current in the same direction (-x axis).

- We can write the diffusion current as (negative sign):

$$J_{px,dif} = -qD_p \frac{dp}{dx}$$

where D_p is the hole diffusion coefficient

Current Components

Four independent current flow mechanism

1. Electron Drift
2. Hole Drift
3. Electron Diffusion
4. Hole Diffusion

For 1-dimensional flow:

- Current density is:

$$J = qn\mu_n E_x + qp\mu_p E_x + qD_n \frac{dn}{dx} - qD_p \frac{dp}{dx}$$

- The mobility and diffusion coefficients are not independent quantities
- In most situations, we need not consider all four mechanisms

Energy Band Diagram and Voltage (1)

Convention: The energy band diagram is drawn in terms of ***energy of the electron***

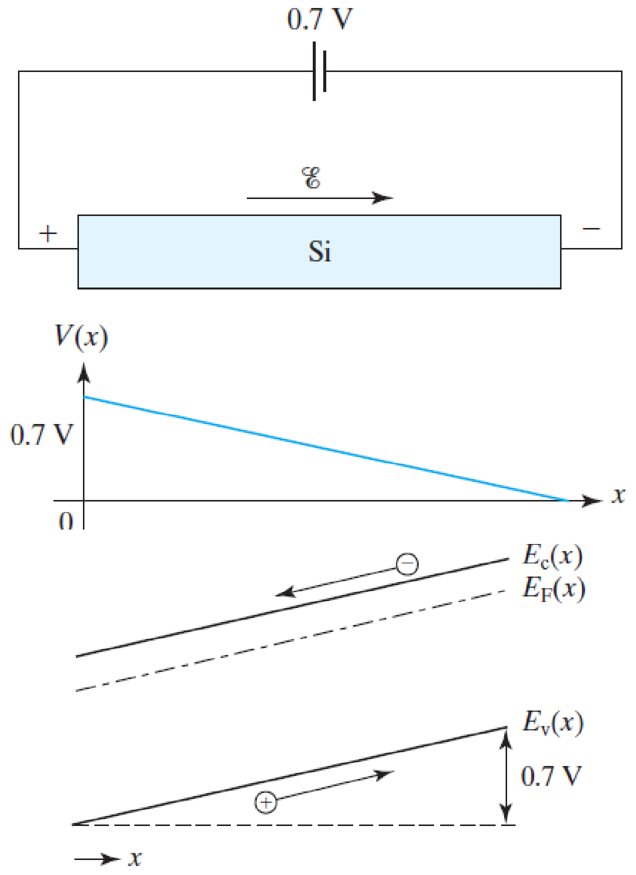
By definition: A positive voltage raises the potential energy of a positive charge and lowers the potential energy of a negative charge.

⇒ The positive voltage will increase the potential energy of holes and reduce the potential energy of electrons

Problem: We apply a voltage of 0.7 V across a silicon bar as shown in the figure.

Draw the

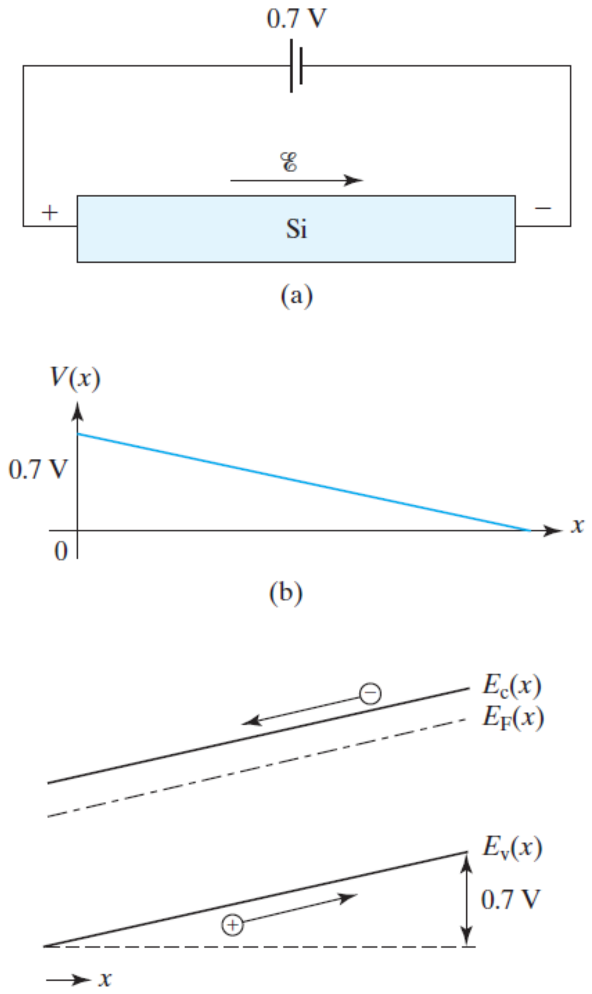
a. Voltage profile as a function of position
b. Energy band diagram



Note:

- E_c and E_V are separated by bandgap
- where voltage is high, E_c and E_V are low
- where voltage is low, E_c and E_V are high

Energy Band Diagram and Voltage (2)



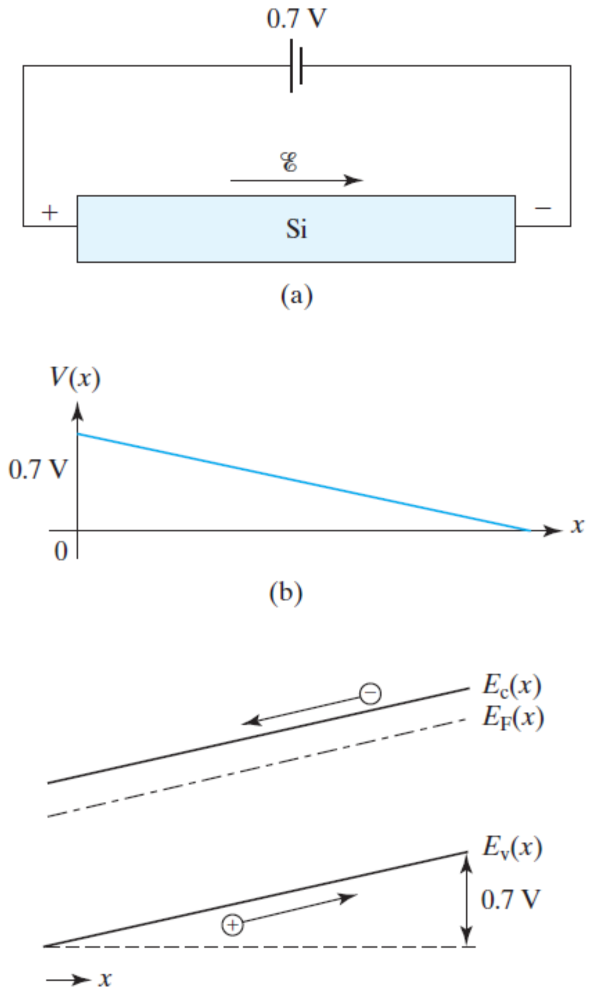
- E_c and E_V vary in opposite direction from the voltage (or electrical potential)

- In general, we can write:

$$E_c(x) = E_{Reference} - qV(x)$$

  - Both E_c and voltage V are considered a function of position
  - $E_{Reference}$ is an arbitrary reference energy
  - The factor q needs to be taken because energy band diagram is shown in eV and voltage is taken in $Volts$, and to convert $Volts$ to eV we need to multiply by q

Energy Band Diagram and Electric Field



- We can write:

$$E_c(x) = E_{Reference} - qV(x)$$

- The electric field is given as:

$$E(x) = -\frac{dV(x)}{dx}$$

$$E(x) = -\frac{d\left(E_{Reference} - E_c(x)\right)/q}{dx}$$

$$\Rightarrow E(x) = \frac{1}{q}\frac{dE_c(x)}{dx}$$

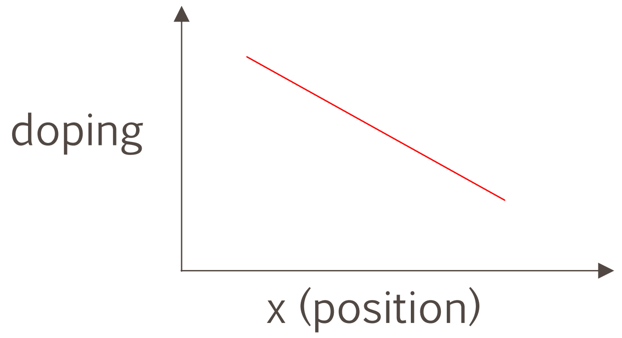
We can also write:

$$\boldsymbol{E(x) = \frac{1}{q}\frac{dE_c(x)}{dx} = \frac{1}{q}\frac{dE_V(x)}{dx}}$$

[Remember this relation]

Induced Electric Field

Consider an n-type semiconductor in which doping decreases along the X-axis. How will diffusion occur and when will it stop (equilibrium be reached)?



- Electrons will diffuse from high concentration region to low concentration region (along the +X-axis) **[Diffusion current]**
- It will leave more +ve ions on the left-hand side (-X axis)
  - Ionized atoms do not move
- An electric field will be induced by separation of charge (along +X axis)

- The induced electric field will apply force on electron along –X axis **[induced electric field force**] and will lead to **Drift current**
- The **diffusion current** and **drift current** will balance and no current will flow when the semiconductor is in equilibrium

The Einstein Relation (1)

Consider a piece of N-type semiconductor in which the dopant density decreases towards right.

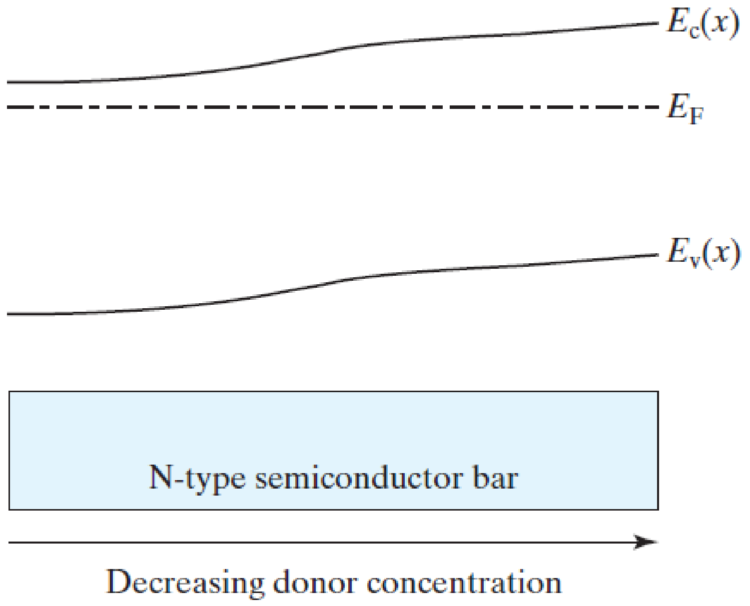
In equilibrium, the Fermi Energy level will be constant

- There is a slope in E_c

  $\Rightarrow$there is an electric field given by $\frac{1}{q}\frac{dE_C(x)}{dx}$

- The electric field will lead to drift current

- Due to gradient in concentration, there will be diffusion current



- At equilibrium these two current will balance each other and no current will flow

- By equating these two current components, we can derive Einstein's relation

The Einstein Relation (2)

Problem: Consider an n-type semiconductor in which doping is given as $N_D(x)$. Assume semiconductor to be non-degenerate.

Assume that the electron concentration is same as doping concentration, i.e., $n(x) = N_D(x)$.

a) Express $n(x)$ as a function of $E_c(x)$ and E_F.

b) Compute the electric field as function of x

c) Compute the drift component of current density

d) Compute the diffusion component of current density

e) Equate the magnitude of two current densities and derive Einstein's relation.

a) $n(x) = N_C exp\left(-\frac{E_C(x)-E_F}{kT}\right)$

b) $E_c(x) = E_F - kTln\left(\frac{n(x)}{N_C}\right)$

$$E(x) = \frac{1}{q}\frac{dE_c(x)}{dx} = -\frac{kT}{q}\frac{1}{n(x)}\frac{dn(x)}{dx}$$

c) $J_{drift}(x) = qn(x)\mu_n E(x)$

$$= -\cancel{qn(x)}\mu_n \frac{kT}{\cancel{q}}\frac{1}{\cancel{n(x)}}\frac{dn(x)}{dx} = kT\mu_n\frac{dn(x)}{dx}$$

d) $J_{diff}(x) = qD_n\frac{dn(x)}{dx}$

e) $kT\mu_n\frac{\cancel{dn(x)}}{\cancel{dx}} = qD_n\frac{\cancel{dn(x)}}{\cancel{dx}} \Rightarrow \frac{D_n}{\mu_n} = \frac{kT}{q}$

The Einstein Relation (3)

Einstein relation for electrons:

$$\frac{D_n}{\mu_n} = \frac{kT}{q}$$

Similarly, we can derive Einstein relation for holes:

$$\frac{D_p}{\mu_p} = \frac{kT}{q}$$

Significance:

- The diffusion coefficient and mobilities are not independent
- The scattering processes that impact carrier drift, also impact carrier diffusion
- Similar to mobility, the diffusion coefficient will be strongly influenced by temperature

Einstein's Relation: Problem

Determine the diffusion coefficient given the electron mobility is 1000 cm^2/Vs at 300 K.

Einstein relation for electrons:

$$\frac{D_n}{\mu_n} = \frac{kT}{q}$$

$$D_n = 0.0259 \times 1000$$

$$= 25.9 cm^2/s$$

Quasi Neutral Condition

- An electric field will be induced by separation of charge (along +X axis)
- How can we assume electron concentration is same as the doping concentration, i.e. $n(x) = N_D(x)$?

Quasi Neutral Assumption: If the dopant concentration profile varies gradually with position, then the majority-carrier concentration distribution does not differ much from the dopant concentration distribution.

Reason:

- Very small charge imbalance is sufficient to create the internal electric field of magnitude normally encountered in devices.
- For such small charge imbalance we can assume $n(x) = N_D(x)$.

Non-equilibrium Excess Carriers

Non-equilibrium Behavior

- When will a semiconductor go into non-equilibrium?
  - Examples:
    - ✓ when we apply a voltage
    - ✓ when we shine light on a semiconductor
    - ✓ sudden change in temperature
- Excess carriers can be generated
- How these excess carriers evolve over time and space are of interest in solid-state devices

Generation and Recombination (Under Equilibrium)

Generation: creation of electron-hole pair

Recombination: annihilation of electron-hole pair

Even at thermal equilibrium

- Generation takes place by jumping from valence band to the conduction band
- Recombination takes place when free electron (in conduction band) fall to the valence band and annihilate the hole
- At equilibrium,
  - no change in carrier concentration,
  - $\Rightarrow$ Generation rate is equal to recombination rate

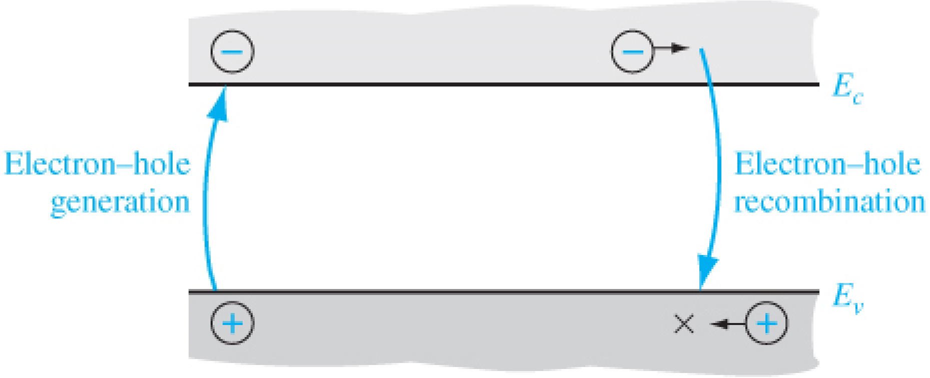


Figure 6.1 | Electron–hole generation and recombination.

G_{no} and G_{po} : generation rates of electrons and holes [unit is $\frac{number}{cm^3 s}$]

R_{no} and R_{po} : recombination rates of electrons and holes [unit is $\frac{number}{cm^3 s}$]

At thermal equilibrium:

$$G_{no} = G_{po} = R_{no} = R_{po}$$

Generation and Recombination (Under Non-equilibrium) (1)

- Excess carriers can be generated by gaining energy from photon (light)
- We may write:

$$n = n_o + \delta n$$
$$p = p_o + \delta p$$

n_o and p_o are equilibrium carrier concentration

δn and δp are excess carrier concentrations

- In non-equilibrium case: $np \neq n_o p_o = n_i^2$

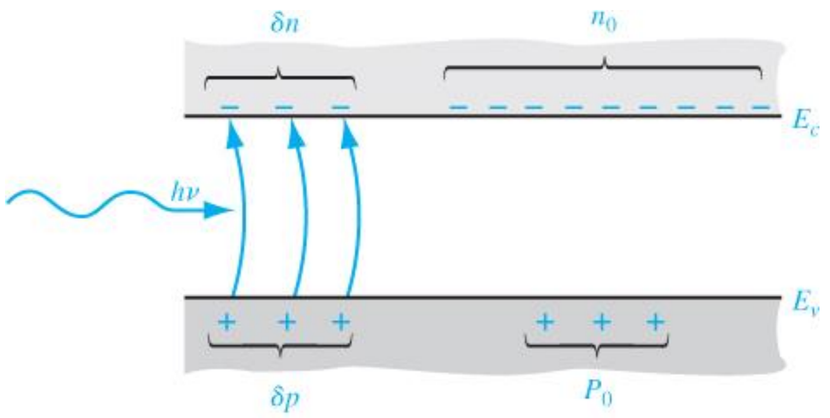


Figure 6.2 | Creation of excess electron and hole densities by photons.

Generation and Recombination (Under Non-equilibrium)

- In the presence of excess carrier we write:

$$n = n_o + \delta n$$
$$p = p_o + \delta p$$

n_o and p_o are equilibrium carrier concentration

δn and δp are excess carrier concentrations

- **Charge neutrality** (electrons and holes are created in pairs):

$$\delta n = \delta p$$

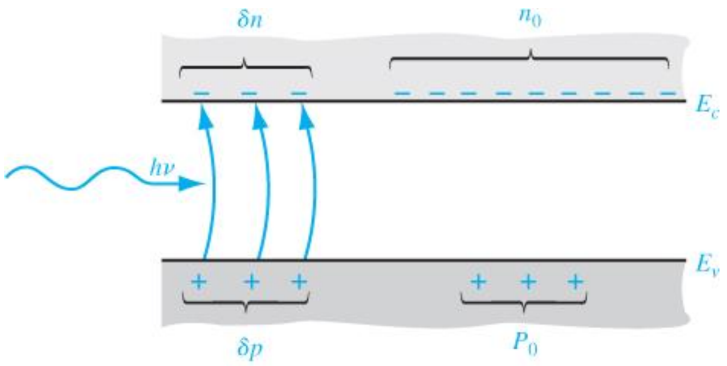


Figure 6.2 | Creation of excess electron and hole densities by photons.

Restoring Equilibrium

- At equilibrium $n_o p_o = n_i^2$

- In non-equilibrium case: $np \neq n_o p_o = n_i^2$

Surplus $(np > n_i^2)$

- Recombination more than generation (increased recombination tries to bring the material back to equilibrium to get rid of surplus)

Deficit $(np < n_i^2)$

- Generation more than recombination (increased generation tries to bring the material back to equilibrium to get rid of deficit)

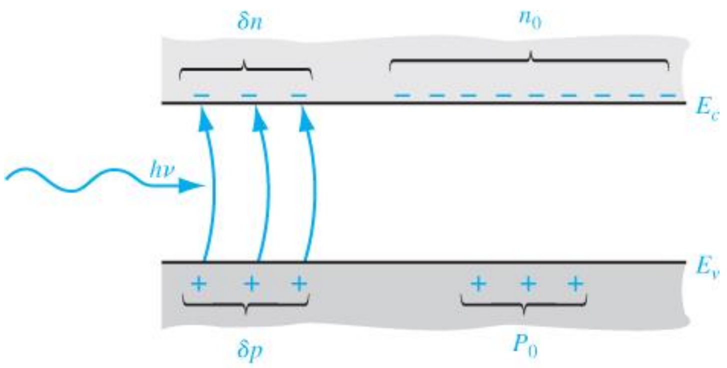


Figure 6.2 | Creation of excess electron and hole densities by photons.

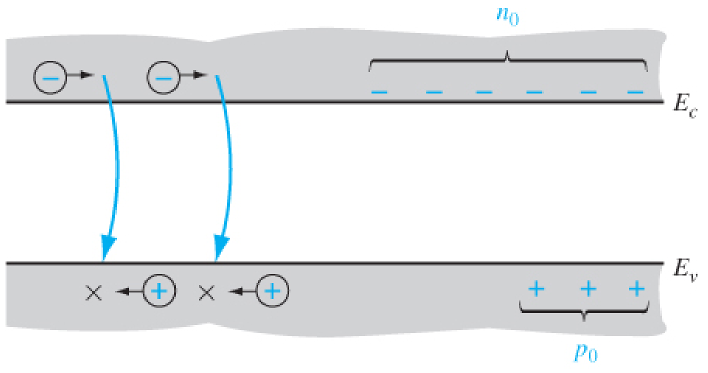


Figure 6.3 | Recombination of excess carriers reestablishing thermal equilibrium.

Rate of change of carrier concentration

- Recombination rate will be proportional to the number of electrons and holes present

$$R \propto np$$
$$R = \alpha_r np$$

- At thermal equilibrium,

$$R = \alpha_r n_0 p_0 = \alpha_r n_i^2$$

- This will be same as spontaneous thermal generation rate:

$$G_{no} = G_{po} = R_{no} = R_{po} = \alpha_r n_i^2$$

- When excess carriers are generated, the net rate in change in electron concentration:

$$\frac{dn}{dt} = \alpha_r n_i^2 - \alpha_r np$$
$$= \alpha_r\left[n_i^2 - (n_o + \delta n)(p_o + \delta p)\right]$$

$$\frac{dn}{dt} = \alpha_r\left[n_i^2 - n_o p_o - n_o \delta p - p_o \delta n - \delta n \delta p\right]$$
$$\frac{dn}{dt} = -\alpha_r[n_o \delta p + p_o \delta n + \delta n \delta p]$$

By charge neutrality (i.e., $\delta n = \delta p$)

$$\frac{dn}{dt} = -\alpha_r[n_o \delta n + p_o \delta n + (\delta n)^2]$$
$$\frac{dn}{dt} = -\alpha_r \delta n[(n_o + p_o) + \delta n]$$

- This equation is difficult to solve.
- But can be solved easily when we assume *low-level injection*

References

- D. A. Neaman, *"Semiconductor Physics and Devices"*